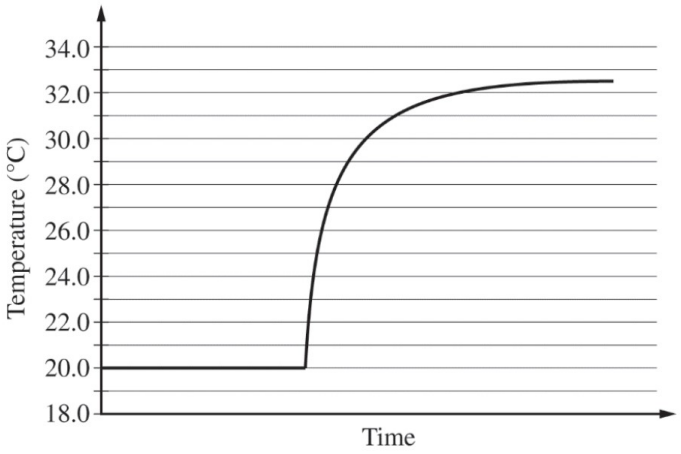
Topics 6.1 – 6.5: MCQ Practice



1. A student initiates a chemical reaction by combining two different solutions, initially at 20.0°C, in an insulated calorimeter. The temperature of the reaction mixture is monitored, as shown in the graph above. Which of the following correctly classifies the energy change that occurs in this reaction and describes the flow of thermal energy between the reacting species (system) and the surroundings?

	Reaction	Flow of Thermal Energy
(A)	endothermic	Thermal energy lost by the reacting species (system) is gained by the surroundings.
(B)	exothermic	Thermal energy lost by the reacting species (system) is gained by the surroundings.
(C)	endothermic	Thermal energy lost by the surroundings is gained by the reacting species (system).
(D)	exothermic	Thermal energy lost by the surroundings is gained by the reacting species (system).

2. A student dissolves a sample of KI(*s*) into a sample of water in a coffee cup calorimeter. The solid dissolves completely, and the student monitors the temperature of the solution, observing that the temperature decreases from 21.0°C to 18.0°C. Which of the following best helps to explain the energy change that occurred during the dissolution of KI(*s*)?

   (A) When KI(*s*) dissolves in water, ions of $K^+(aq)$ and $I^-(aq)$ are formed.

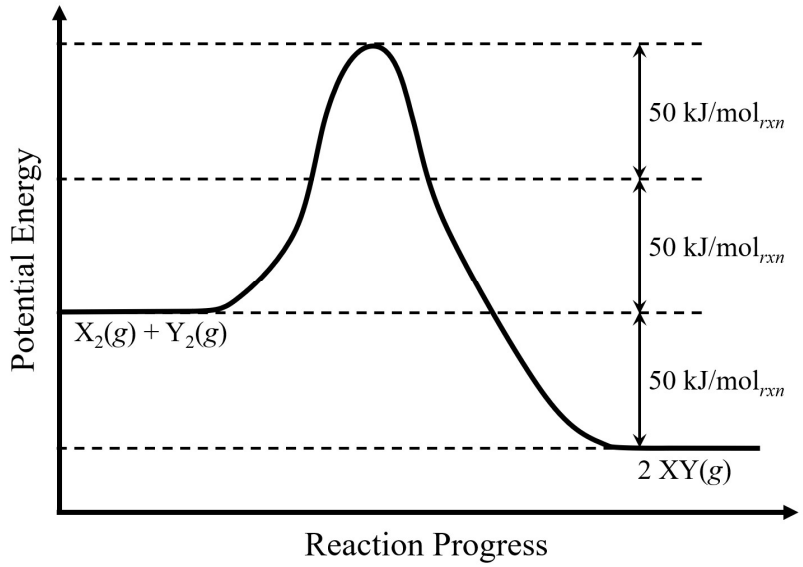
   (B) When KI(*s*) dissolves in water, molecules of KI(*aq*) are formed.

   (C) K^+ ions have very strong ion-ion interactions with I^- ions in the crystal lattice.

   (D) K^+ ions have very strong ion-dipole interactions with water molecules in the solution.

$$X_2(g) + Y_2(g) \rightarrow 2\,XY(g) \qquad \Delta H_{rxn} = ?$$

3. The diagram below represents the energy profile for the chemical reaction in which XY(g) is produced according to the equation above. Which of the following correctly classifies the energy change that occurs in this reaction and gives the correct value for ΔH_{rxn} ?



	Reaction	ΔH_{rxn} (kJ/mol$_{rxn}$)
(A)	endothermic	+50
(B)	exothermic	–50
(C)	endothermic	+100
(D)	exothermic	–100

Substance	Specific Heat (J/(g·°C))
Cd(s)	0.23
Zn(s)	0.39

4. Based on the information in the table above, which of the following experiments absorbs the greatest quantity of thermal energy?

(A) Increasing the temperature of 4.0 g of Cd(s) from 20°C to 90°C

(B) Increasing the temperature of 4.0 g of Zn(s) from 20°C to 90°C

(C) Increasing the temperature of 6.0 g of Cd(s) from 20°C to 60°C

(D) Increasing the temperature of 6.0 g of Zn(s) from 20°C to 60°C

5. A 100 g sample of a metal was heated to 100°C and then quickly transferred to an insulated container holding 100 g of water at 22°C. The temperature of the water rose to reach a final temperature of 35°C. Which of the following can be concluded?

(A) The metal temperature changed more than the water temperature did; therefore the metal lost more thermal energy than the water gained.

(B) The metal temperature changed more than the water temperature did, but the metal lost the same amount of thermal energy as the water gained.

(C) The metal temperature changed more than the water temperature did; therefore the heat capacity of the metal must be greater than the heat capacity of the water.

(D) The final temperature is less than the average starting temperature of the metal and the water; therefore the total energy of the metal and water decreased.

Substance	Specific Heat Capacity (J/(g·°C))	Mass of Sample (g)	Initial Temperature (°C)	Final Temperature (°C)
$CH_3OH(l)$	2.14	20.0	25.0°C	55.0°C
$H_2O(l)$	4.18	20.0	25.0°C	55.0°C

6. A student heated samples of $CH_3OH(l)$ and $H_2O(l)$. Data from the experiment is shown in the table above. Based on the information in the table, which of the following correctly compares the change in the average kinetic energy of the molecules in the samples and provides the correct justification?

	Change in the Average Kinetic Energy of the Molecules	Justification
(A)	higher for the sample of $CH_3OH(l)$	$CH_3OH(l)$ has a lower specific heat capacity and absorbs more energy
(B)	higher for the sample of $H_2O(l)$	$H_2O(l)$ has a higher specific heat capacity and absorbs more energy
(C)	the same for both samples	Each sample has the same mass.
(D)	the same for both samples	The initial and final temperatures are the same.

$$MgO(s) + 2\,HCl(aq) \rightarrow MgCl_2(aq) + H_2O(l) \qquad \Delta H = ?$$

7. A student performed an experiment to determine the value of the enthalpy change, ΔH, for the reaction between MgO(*s*) and HCl(*aq*) as represented by the equation shown above. The student combined a sample of MgO(*s*) with an excess amount of HCl(*aq*) in a coffee-cup calorimeter. Data from the experiment are shown in the table below.

Mass of MgO(*s*) (molar mass = 40.3 g/mol)	4.03 g
Total Mass of the Reaction Mixture	100.00 g
Initial Temperature of Reaction Mixture	20.0°C
Final Temperature of Reaction Mixture	55.7°C
Specific Heat Capacity of the Reaction Mixture	4.2 J/(g·°C)

Which of the following is the enthalpy change of the reaction represented above?

(A) –150 kJ/mol$_{rxn}$

(B) –15 kJ/mol$_{rxn}$

(C) +15 kJ/mol$_{rxn}$

(D) +150 kJ/mol$_{rxn}$

	Trial 1	Trial 2
Mass of LiCl(*s*)	10.0 g	10.0 g
Mass of $H_2O(l)$	90.0 g	190.0 g
Initial Temperature of $H_2O(l)$	20.0°C	20.0°C
Final Temperature of Solution	40.8°C	?

8. A student performs a calorimetry experiment to determine the change in enthalpy for the dissolution of LiCl(*s*) in water. Data from the experiment is shown in the table above. Which of the following is the most likely value for the final temperature of the solution in Trial 2?

(A) 20.4°C

(B) 30.4°C

(C) 40.8°C

(D) 61.6°C

$$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$$

Trial	Volume of 0.10 *M* HCl(*aq*)	Volume of 0.10 *M* NaOH(*aq*)	Amount of Heat Released
1	50. mL	50. mL	X
2	100. mL	50. mL	Y

9. A student conducted an experiment to determine ΔH_{rxn} for the reaction between HCl(*aq*) and NaOH(*aq*). The student ran two trials using the volumes of HCl(*aq*) and NaOH(*aq*) indicated in the table above, and determined the amount of heat released. Which of the following best explains the relationship between X and Y?

   (A) Y = 2X, because the volume of HCl(*aq*) used in trial 2 is twice the volume used in trial 1.

   (B) Y = X, because the number of moles of acid and base reacting with each other is the same in both trials.

   (C) Y = $\frac{2X}{3}$, because the heat is distributed over more particles in trial 2 than in trial 1.

   (D) The relationship between X and Y cannot be predicted.

Data for Gallium (Ga)	
Specific heat capacity of Ga(*l*)	0.37 J/(g·°C)
Heat of fusion	80. J/g
Melting Point	29.8°C

10. A sample of Ga(*s*) at 29.8°C is added to 100. g of Ga(*l*) at 83.9°C. The mixture is stirred gently until the temperature is 29.8°C. All of the remaining Ga(*s*) is quickly removed. Based on the information in the table above, the mass of Ga(*s*) that melted in this experiment is closest to

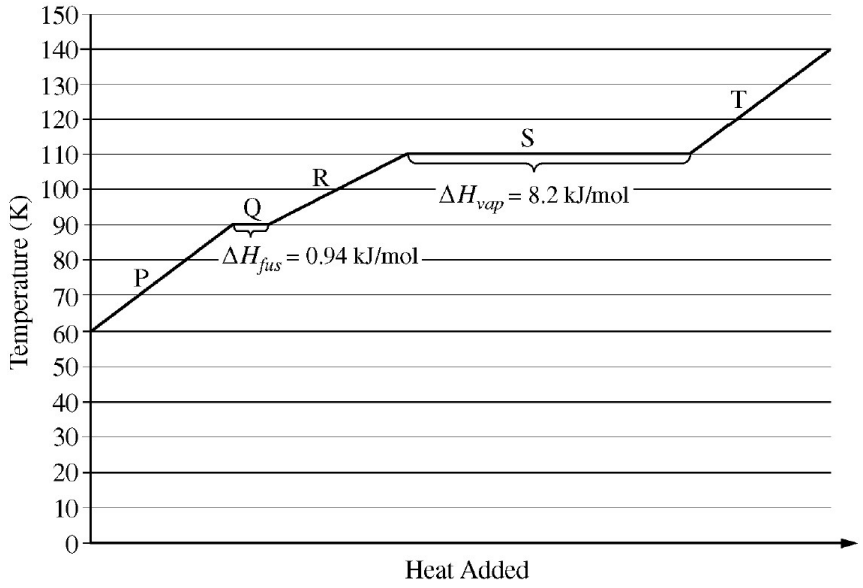
    (A) 0.25 g

    (B) 14 g

    (C) 25 g

    (D) 39 g

11. The heating curve for methane, CH_4, is shown below.



The energy required to vaporize a 120 g sample of $CH_4(l)$ at its boiling point is closest to

(A) 1.1 kJ

(B) 7.1 kJ

(C) 62 kJ

(D) 980 kJ